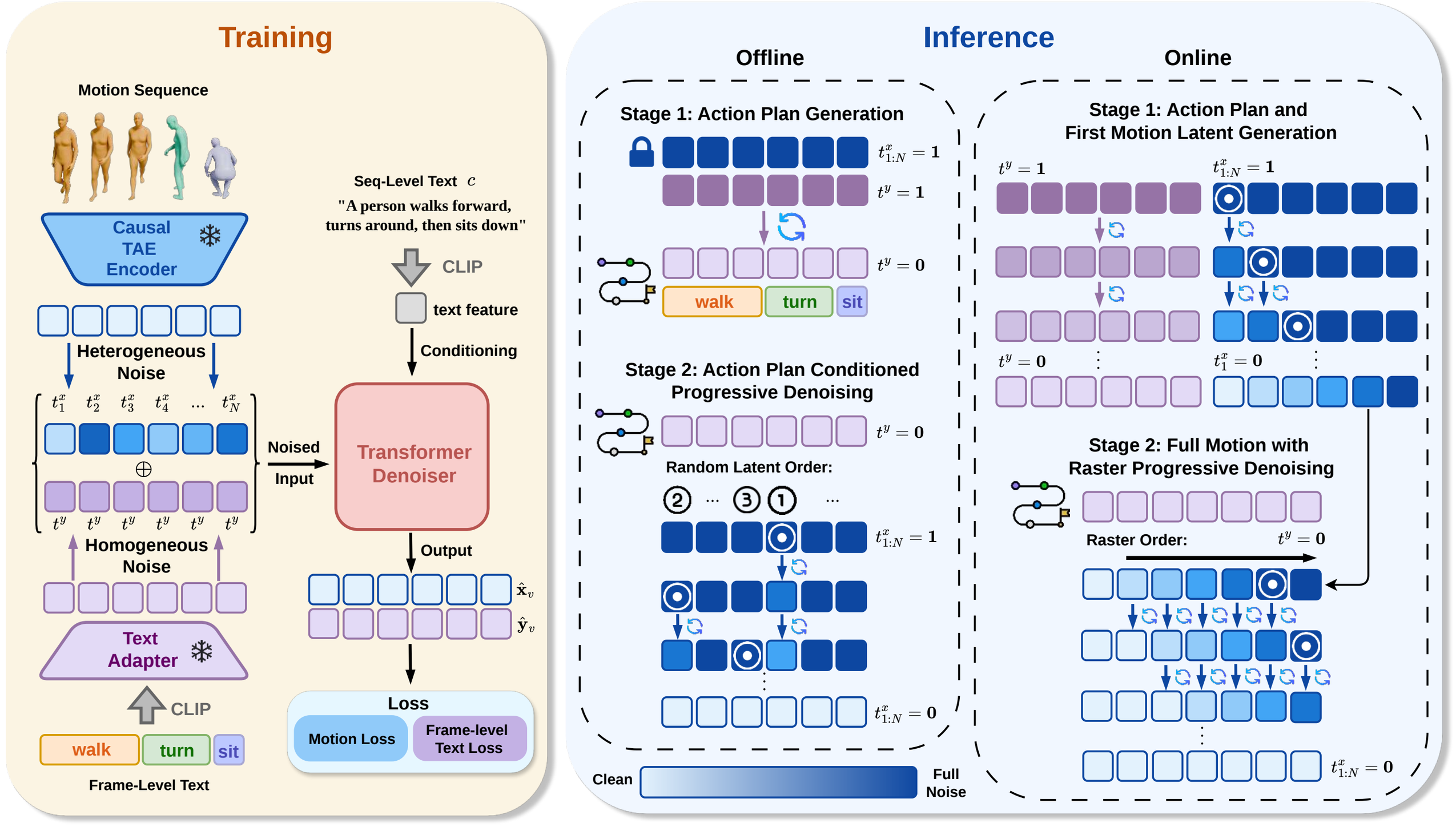
Training
Motion Sequence
Causal TAE Encoder
Heterogeneous Noise
t_1^x t_2^x t_3^x t_4^x ... t_N^x
t^y t^y t^y t^y t^y t^y
Homogeneous Noise
Text Adapter
CLIP
walk
turn
sit
Frame-Level Text
Seq-Level Text c
"A person walks forward, turns around, then sits down"
CLIP
text feature
Conditioning
Noised Input
Transformer Denoiser
Output
$\hat{\mathbf{x}}_v$
$\hat{\mathbf{y}}_v$
Loss
Motion Loss
Frame-level Text Loss
Inference
Offline
Stage 1: Action Plan Generation
$t_{1:N}^x = 1$
$t^y = 1$
$t^y = 0$
walk
turn
sit
Stage 2: Action Plan Conditioned Progressive Denoising
$t^y = 0$
Random Latent Order:
2 ... 3 1 ...
$t_{1:N}^x = 1$
$t_{1:N}^x = 0$
Online
Stage 1: Action Plan and First Motion Latent Generation
$t^y = 1$
$t_{1:N}^x = 1$
$t^y = 0$
$t_1^x = 0$
Stage 2: Full Motion with Raster Progressive Denoising
Raster Order:
$t^y = 0$
$t_{1:N}^x = 0$
Clean
Full Noise